Research Program: The Spectral Core (V01–V19, Th1–Th6, V13)
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RESEARCH PROGRAM · CORE
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Research Program: The Spectral Core

The core program owns the mathematical foundation of the AB architecture: the blind spectrum, null models, the critical line, the trace formula, the obligation machine and branch stitching. Nineteen campaigns and six theorems form a closed loop: construction → certificate → formal statement → independent verification.

1. Program goals

- construct a Hermitian operator H_ξ whose heat expansion carries the arithmetic of the primes ($\Lambda(n)/\sqrt{n}$);
- prove the correctness of the construction: Hermiticity (Th1), self-adjointness (Th2), Kato certificates (Th3), the Li formula (Th4), block reduction (Th5), the Guinand–Weil correspondence (Th6);
- separate arithmetic specificity from GUE statistics — including honest negative answers on lattices;
- fix the RH status with the obligation machine V13 — and keep it blocked while the hypotheses are unproved.

2. Methodology

Every campaign starts with a pre-registration protocol: thresholds are fixed before the run. Blind protocols are applied wherever a reference exists (V01: calibration/test with no leak). All derived artifacts (tiles, documents) are built only from committed verdicts. Formal statements are duplicated in Lean 4, Coq, Agda and Isabelle (reductions A1–A6, no sorry/admit).

3. Executed campaigns

p0.10 p0.29 p0.58 **Campaign** & **Content** & **Key result** V01 & blind spectrum & statistics consistent; individual ordinates not predicted (honest negative)
V02 & null models & AB cloud in the GUE class; specificity measured by fine structure
V03 & lattice σ-scan & no agreement zone with ζ found
V04 & Hermiticity & Th1: $\|H - H^\dagger\| = 0.0$; symmetry $\lambda \leftrightarrow -\lambda$ at $\alpha = ½$
V05 & construction certificate & construction boundaries fixed

V06 & level identity & gates at the zeros' level — YES (the D_nn bridge, v2.0.7); bit-level branch open
V07 & trace ↔ primes & Th6: r = 0.9996; lattice control 0.76; the B/C_0 erratum
V08 & self-adjointness & Th2: defects (0,0); unitarity drift 2.2e−16
V09 & block reduction & Th5: block union ≈ spectrum to 7.1e−15
V10 & spectral certificates & certificates consistent with Kato estimates
V11 & obligation DAG draft & the structure of the path to RH fixed
V12 & Kato certificates & Th3: widths ~1e−27 (dps 40)
V13 & obligation machine & RH blocked: 5 proven / 3 computed / 4 assumed
V14 & claim audit & claim files consistent with verdicts
V15 & Li formula & Th4: two paths ≤ 4.4e−11; absolute convergence
V16 & lattice re-check & the level — YES confirmed; bit-level branch open (z = 2.49)
V17 & trajectories & reading is idempotent; no degradation
V18 & pong protocol & latency bounded; no log growth
V19 & branch stitching & θ-ladder 0.832; coherence 0.9995; Z_2-monodromy

4. The next wave

- stitching refinement: r_align ≤ 0.135 on an extended height grid;
- the ζ monodromy as an algebraic object: formalization of V19;
- compiling obligations into Lean/Coq: O-lean/O-coq — moving the V13 DAG into the formal layer;
- extending the Th3–Th4 certificates to larger sizes.

5. Honest boundaries

Individual ordinates are not predicted by the blind spectrum — statistics agreed, no oracle. The hypotheses O08/O09/O11/O12 are not removed: RH stays blocked. Vortex terms break Th5. Lattice constructions do not reproduce the arithmetic of ζ. These boundaries are part of the result, not its flaw.

6. Key campaigns in detail

6.1. Six campaigns that defined the core

V01 — the blind spectrum

The campaign is built as a strict no-leak check: the operator is calibrated on 600 ordinates and tested on another 600. The result is dual and both components matter. Positive: the test-set statistics agree with the construction expectations — correlations within control zones. Negative:

individual ordinates are not predicted. The practical conclusion: the model works as a consistency counter, not as an oracle; any future claim of "zero prediction" must first refute this verdict.

V02 — null models and GUE

The comparison with four null models (M0, N1, N4 and references) over r_mean, Σ_2, Δ_3 and the KS statistic showed: the AB cloud falls into the GUE class together with the controls (r_mean 0.6093 versus 0.5837 for random phases, KS 0.0499). Hence the program's methodological rule: GUE agreement is necessary but not sufficient; the arithmetic specificity is sought separately — in the trace, the constants and the behaviour.

V07 — trace ↔ primes

The central arithmetic campaign of the core. The heat trace K(x) = Tr e^(−xH) expands in a series whose coefficients carry $\Lambda(n)/\sqrt{n}$ — the content of Th6. The trace-prime correlation r = 0.9996; the lattice control without arithmetic gives 0.76; the gap is the measured specificity. An important side result — the B/C_0 constants erratum: the error was found by an independent derivation and fixed in the same verdict.

V12 — Kato certificates

The campaign answers the question "how much can a computed eigenvalue be trusted". Kato theory gives the corridor δ^2/gap plus a Gershgorin allowance; numerically the widths are of order 1e−27 at forty significant digits. A rare situation: the error of the estimate is known better than the estimate itself. These certificates later become the basis of the computed status in the obligation machine.

V13 — the obligation machine

The executable DAG of the path to RH: five nodes proven, three computed, four assumed (O08, O09, O11, O12); the conclusion is automatically blocked. The verdict note fixes the principle: no numerical check raises the status. The campaign turned the program from a set of results into a system of obligations with checkable statuses.

V19 — branch stitching

The final algebraic campaign of the core: the branches of the spectral construction are stitched with a θ-ladder, agreement 0.832 and coherence 0.9995 (z = 43.4); Z_2-monodromy. The certificate is algebraic: going around the branch point twice returns to the original branch. The stitching closes the construction and opens the R2 roadmap: refining r_align ≤ 0.135 and formalizing the monodromy.

7. The full campaign table

p0.12 p0.25 p0.60 **Campaign** & **Title** & **Key result** V1 & Blind spectrum & слепой контроль выполнен; индивидуальные ординаты не предсказываются (r_test в нулевой зон
V2 & Null models & AB-cloud попадает в GUE-класс вместе с N1/N4; специфичность арифметики измеряется не ⟨r⟩,
V3 & Lattice scan & минимум D(σ) в окрестности σ = 1/2 отсутствует или неустойчив — специфичность линии измере
V4 & Hermiticity & Th1 доказана: эрмитовость по построению, спектр вещественен, киральность α=½/W=0 — бипарти
V5 & Construction certificate & устойчивый степенной закон конечномерной ошибки обнаружен; экстраполяция за пределы прогон
V6 & Level identity & тождество уровней (n ↔ нуль n) для решёточных затворов — ДА (мост D_nn, v2.0.7); ранговая ветвь на пределе разрешения
V7 & Trace ↔ primes (Th6) & вес простых Λ(n)/√n доказательно живёт в тепловом следе H_ξ (Th6: |ΔK| ≤ 5e-5); в следах p
V8 & Self-adjointness & Th2 доказана для конечной размерности; устойчивость предельной модели при смене BC — измер
V9 & Block reduction & Th5 доказана и верифицирована (тор без вихрей: спектр = объединение блоков характеров, вне
V10 & Spectral certificates & часть арифметики (заряды) идентифицируема по старшим следам при достаточном k; ранг J < n_
V11 & Obligation DAG draft & GUE-универсальность выживает при всех режимах разрушения арифметики (⟨r⟩ в классе); арифме
V12 & Kato certificates & все сертификаты содержат истинные значения; интервалы — гарантированные, а не десятичные д
V13 & Obligation machine & формальная машина БЛОКИРУЕТ вывод RH: путь к заключению содержит гипотезы O08/O09/O11/O12;
V14 & Claim audit & каждое публичное утверждение пакета трассируется к вычислению, контролю и перечню не-следс
V15 & Li formula & Th4 доказана и верифицирована (два пути, |Δ| ≤ 1e-9); λ_n > 0 для n ≤ 3000 измерено с серт
V16 & Lattice re-check &
V17 & Trajectories &
V18 & Pong protocol &
V19 & Branch stitching &
V20 & Processor compilation (P0) & 1−F ≤ 4.4e-16; глубина ≤ 7/29

8. How to reproduce

```
cd modules/core_spectrum
python3 -m pytest tests/ -q            # core tests
python3 scripts/compute_all.py         # rebuild computations
```

```
python3 scripts/make_figures.py          # level figures
# web lab: modules/core_spectrum/web_lab/index.html
```

9. Further reading

- core docs: modules/core_spectrum/docs/ — theory and methods;
- level monographs: modules/core_spectrum/monograph/ (incl. appendix_volume);
- verdicts: modules/core_spectrum/data/v01…v19_verdict.json;
- formal statements: formal/ — reductions A1–A6.